
NETWORK SCANNING AND ENUMERATION USING NMAP

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Tools Used

- Nmap
- Enum4linux
- SMBClient
- Nslookup , Whois
- Weplizer , Builtwith
- Kali Linux
- Metasploitable 2

Abstract

This project demonstrates the practical implementation of network reconnaissance and enumeration techniques using Nmap and supporting tools. The objective was to identify active hosts, discover open ports, enumerate services, perform operating system detection, and gather information about SMB shares in a controlled laboratory environment.

1. Introduction

Network scanning is one of the first phases of penetration testing and security assessment. It helps security professionals identify active systems, open ports, running services, and possible attack surfaces within a network.

In this project, Nmap was used to perform various scanning techniques and enumeration tasks against laboratory targets.

2. Objectives

- Discover live hosts on the network
 - Identify open TCP ports
 - Detect running services
 - Perform OS fingerprinting
 - Enumerate SMB resources
 - Understand reconnaissance methodologies
-

3. Lab Setup

Component	Description
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2 / Windows
Network Type	Local Virtual Network
Scanning Tool	Nmap
Enumeration Tool	Enum4linux, SMBClient

4. Methodology

Flow

Target Identification -> Host Discovery -> Port Scanning -> Service Enumeration -> OS Detection -> SMB Enumeration -> Result Analysis

5. Practical Implementation

5.1 Host Discovery

Command

```
nmap -sP 192.168.186.130
```

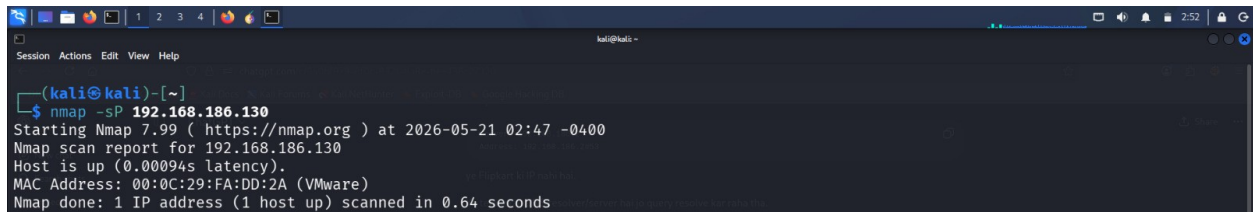
Purpose

To determine whether the target host is active on the network.

Result

- Host was found alive.
- MAC address information was identified.

Screenshot

A screenshot of a Kali Linux terminal window. The terminal shows the command `nmap -sP 192.168.186.130` being executed. The output indicates that the host is up, with a latency of 0.00094s, and provides the MAC address 00:0C:29:FA:DD:2A (VMware). The scan was completed in 0.64 seconds.

```
(kali@kali)~$ nmap -sP 192.168.186.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 02:47 -0400
Nmap scan report for 192.168.186.130
Host is up (0.00094s latency).
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.64 seconds
```

5.2 DNS Resolution

Command

```
nslookup flipkart.com
whois www.example.com -> Registrar Info
```

Purpose

To resolve a domain name into an IP address. and ip into domain name.

Result

- Domain successfully resolved.
- Public IP address obtained.

Screenshot

```
(kali㉿kali)-[~]
$ nslookup flipkart.com
Server:      192.168.186.2
Address:     192.168.186.2#53

Non-authoritative answer:
Name:   flipkart.com
Address: 163.53.76.86
```

5.3 Fast Port Scan

Command

```
nmap -F 163.53.76.86
```

Purpose

To quickly identify commonly used open ports.

Result

- Port 80 detected.
- Port 443 detected.

Screenshot

```
(kali㉿kali)-[~]
$ nmap -F 163.53.76.86

Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 02:51 -0400
Nmap scan report for 163.53.76.86
Host is up (0.0079s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 2.53 seconds
```

5.4 SYN Stealth Scan

Command

```
nmap -sS 163.53.76.86
```

Purpose

To perform a stealth TCP SYN scan.

Result

- Open ports identified.
- Faster and stealthier scanning observed.

Screenshot

```
Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds
(kali@kali)-[~]
$ nmap -sS 163.53.76.86
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 02:55 -0400
Nmap scan report for 163.53.76.86
Host is up (0.0072s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  https
Nmap done: 1 IP address (1 host up) scanned in 46.76 seconds
(kali@kali)-[~]
$
```

5.5 OS Detection

Command

```
nmap -O 163.53.76.86
```

Purpose

To identify the target operating system.

Result

- OS fingerprinting performed.
- Partial operating system information obtained.

```
(kali@kali)-[~]
$ nmap -O 163.53.76.86
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 03:00 -0400
Nmap scan report for 163.53.76.86
Host is up (0.0085s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  https
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.79 seconds
```

5.6 Service Detection

Command

```
nmap -A 163.53.76.86
```

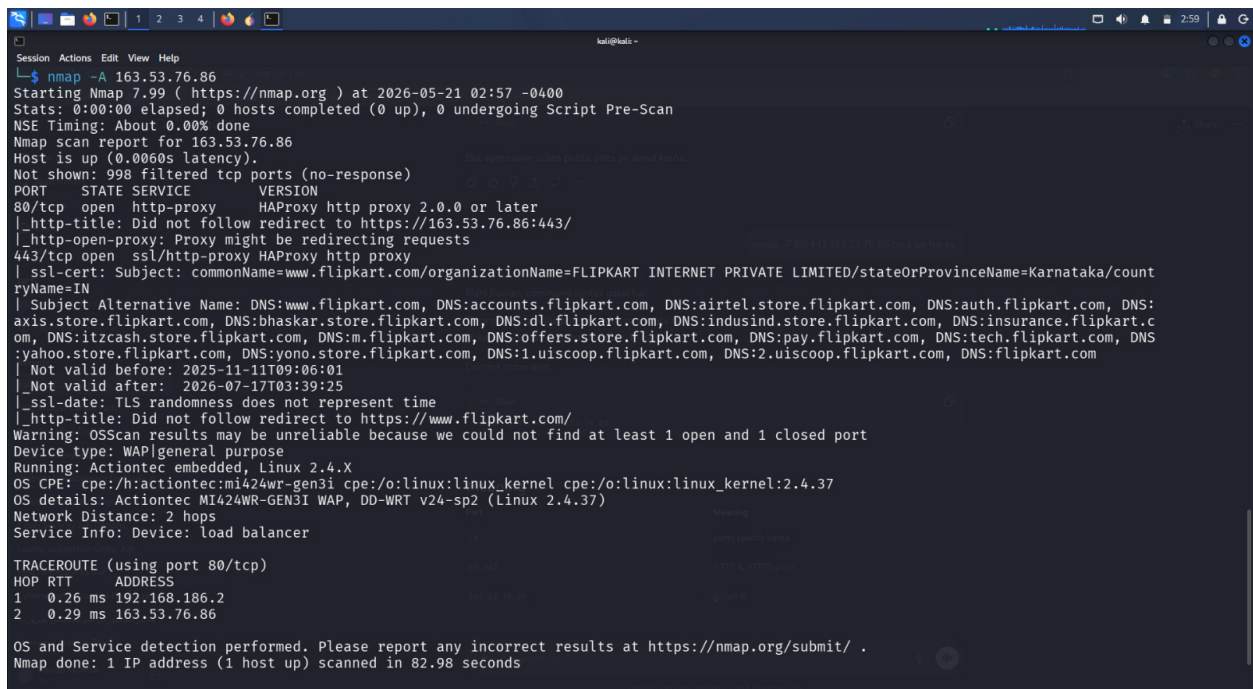
Purpose

To detect service versions and gather additional information.

Result

- HTTP service detected.
- HTTPS service detected.
- Additional service information collected.

Screenshot



```
Session Actions Edit View Help
└─$ nmap -A 163.53.76.86
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 02:57 -0400
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan
NSE Timing: About 0.00% done
Nmap scan report for 163.53.76.86
Host is up (0.0060s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http-proxy   HAProxy http proxy 2.0.0 or later
|_http-title: Did not follow redirect to https://163.53.76.86:443/
|_http-open-proxy: Proxy might be redirecting requests
443/tcp    open  ssl/http-proxy HAProxy http proxy
|_ssl-cert: Subject: commonName=www.flipkart.com/organizationName=FLIPKART INTERNET PRIVATE LIMITED/stateOrProvinceName=Karnataka/count
ryName=IN
|_Subject Alternative Name: DNS:www.flipkart.com, DNS:accounts.flipkart.com, DNS:airtel.store.flipkart.com, DNS:auth.flipkart.com, DNS:
axis.store.flipkart.com, DNS:bhaskar.store.flipkart.com, DNS:dl.flipkart.com, DNS:indusind.store.flipkart.com, DNS:insurance.flipkart.c
om, DNS:itzcash.store.flipkart.com, DNS:m.flipkart.com, DNS:offers.store.flipkart.com, DNS:pay.flipkart.com, DNS:tech.flipkart.com, DNS
:yahoo.store.flipkart.com, DNS:yono.store.flipkart.com, DNS:1.uiscoop.flipkart.com, DNS:2.uiscoop.flipkart.com, DNS:flipkart.com
|_Not valid before: 2025-11-11T09:06:01
|_Not valid after: 2026-07-17T03:39:25
|_ssl-date: TLS randomness does not represent time
|_http-title: Did not follow redirect to https://www.flipkart.com/
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: WAP|general purpose
Running: Actiontec embedded, Linux 2.4.X
OS CPE: cpe:/h:actiontec:mi424wr-gen3i cpe:/o:linux:linux_kernel cpe:/o:linux:linux_kernel:2.4.37
OS details: Actiontec MI424WR-GEN3I WAP, DD-WRT v24-sp2 (Linux 2.4.37)
Network Distance: 2 hops
Service Info: Device: load balancer

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.26 ms 192.168.186.2
2 0.29 ms 163.53.76.86

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 82.98 seconds
```

5.7 SMB Enumeration

Command

```
smbclient -L //192.168.186.130
```

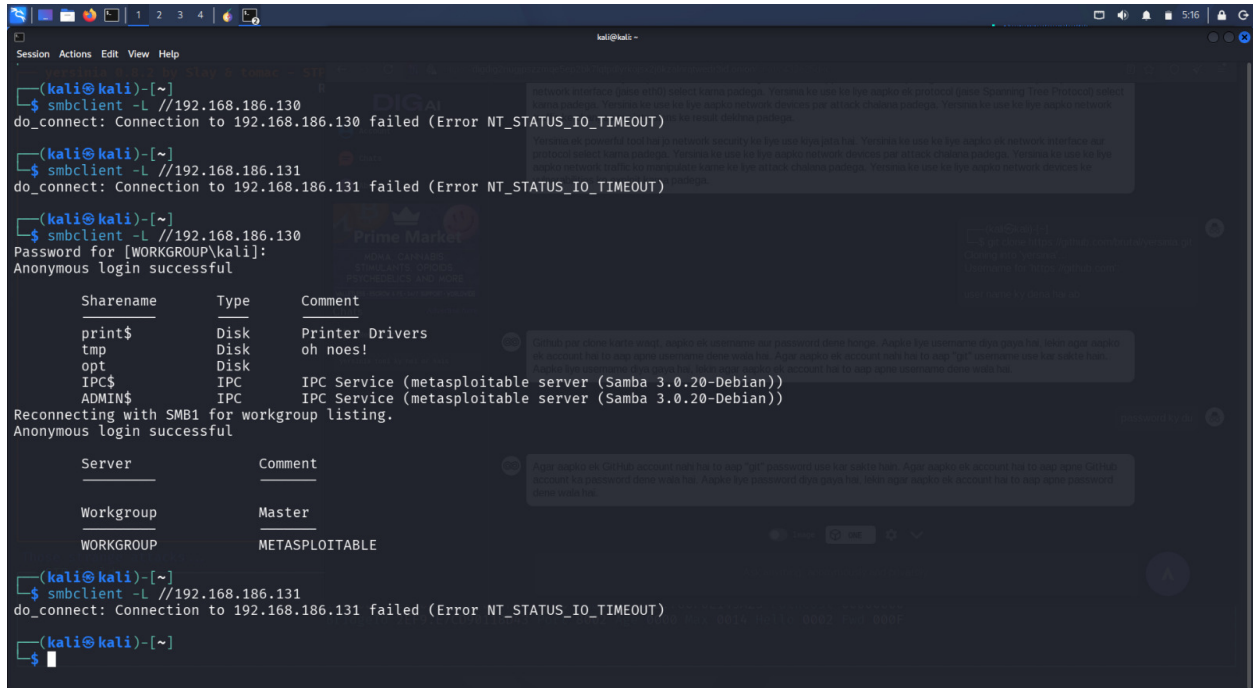
Purpose

To enumerate available SMB shares.

Result

- Shared resources discovered.
- SMB service information collected.

Screenshot



```
(kali@kali)-[~]
$ smbclient -L //192.168.186.130
do_connect: Connection to 192.168.186.130 failed (Error NT_STATUS_IO_TIMEOUT)

(kali@kali)-[~]
$ smbclient -L //192.168.186.131
do_connect: Connection to 192.168.186.131 failed (Error NT_STATUS_IO_TIMEOUT)

(kali@kali)-[~]
$ smbclient -L //192.168.186.130
Password for [WORKGROUP\kali]:
Anonymous login successful

Sharename      Type      Comment
-----
print$         Disk      Printer Drivers
tmp            Disk      oh noes!
opt            Disk
IPC$           IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN$         IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))

Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

Server          Comment
-----
Workgroup       Master
WORKGROUP      METASPLOITABLE

(kali@kali)-[~]
$ smbclient -L //192.168.186.131
do_connect: Connection to 192.168.186.131 failed (Error NT_STATUS_IO_TIMEOUT)

(kali@kali)-[~]
$
```

5.8 Enum4linux Enumeration

Command

```
enum4linux -A 192.168.186.130
```

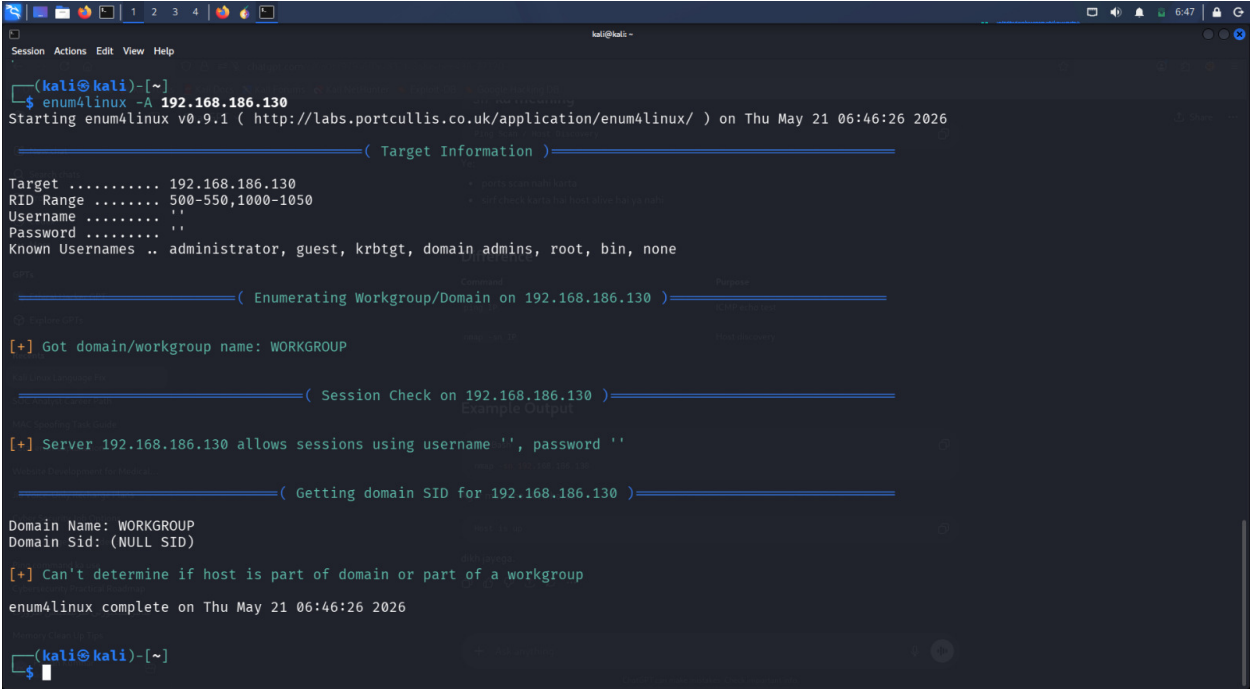
Purpose

To enumerate users, shares, and SMB information.

Result

- Workgroup information identified.
- SMB-related details collected.

Screenshot



6. Findings

Activity	Finding
Host Discovery	Target was active
Port Scanning	Multiple ports identified
Service Detection	HTTP and HTTPS detected
OS Detection	OS fingerprint information gathered
SMB Enumeration	Shares identified
Enum4linux	Workgroup and SMB information obtained

7. Skills Demonstrated

- Network Reconnaissance
 - Host Discovery
 - Port Scanning
 - Service Enumeration
 - OS Fingerprinting
 - SMB Enumeration
 - Linux Command Line Operations
 - Information Gathering
-

8. Commands

- `nmap < Target ip >`
 - `nmap -sP < Target ip >` ping sweep (live host)
 - `nmap -F < Target ip >` Fast scan(100 port)
 - `nmap -P 80,443 < Target Ip >` scan specific port
 - `nmap -sV < Target Ip >` service / version
 - `nmap -sS < Target Ip >` syn scan (undetected by many firewalls)
 - `nmap -sT < Target Ip >` Tcp connect - full connection scan
 - `nmap -A < Target Ip >` aggressive
 - `nmap -o < Target Ip >` os dection
 - `nmap -p - < Target Ip >` scan all port
 - `nmap --open < Target Ip >` scan only open port
 - `nmap -p --open -T 4 < Target Ip >`
-

9. Learning Outcomes

After completing this project, I learned:

- How network reconnaissance works.
- How Nmap scanning techniques differ.
- How to identify services running on a host.
- How SMB enumeration is performed.
- How attackers and defenders use reconnaissance data.

10. Conclusion

This project successfully demonstrated the use of Nmap and related enumeration tools to perform network reconnaissance in a controlled environment. The project provided practical experience in host discovery, port scanning, service detection, operating system fingerprinting, and SMB enumeration.